



Resilience-Oriented Multimodal Transport Planning for Humanitarian Aid Delivery Under Disaster-Induced Infrastructure Disruptions

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Abstract: Natural disasters frequently disrupt transport infrastructure and restrict access to affected communities, making the timely delivery of humanitarian aid a major challenge for disaster management. Selecting a suitable multimodal transport solution requires decision-makers to balance delivery time, cost, infrastructure availability, reliability, safety, flexibility, and capacity. This study investigates the selection of a resilient multimodal transport solution for humanitarian aid delivery under disaster-induced infrastructure disruptions in Bosnia and Herzegovina. An integrated Full Consistency Method (FUCOM) and Measurement of Alternatives and Ranking according to Compromise Solution (MARCOS) model was applied to a case involving the delivery of humanitarian aid from the Armed Forces logistics base at Rajlovac Barracks to Donja Jablanica following the floods and landslides of October 2024. The FUCOM was used to determine the weights of seven evaluation criteria, and the MARCOS method was employed to rank five transport alternatives. Delivery time received the highest weight (0.236), followed by transport system reliability and resilience (0.184) and safety (0.154). The results showed that the integrated road–rail–air alternative achieved the highest utility value (0.7309), narrowly exceeding direct air transport (0.7151). A sensitivity analysis covering 70 weight-variation scenarios showed that the leading alternatives retained stable positions across nearly all scenarios. The findings indicate that integrated multimodal transport can provide a resilient response when individual transport links are disrupted or unavailable. The proposed model offers a transparent decision-support framework for transport planning, infrastructure contingency assessment, and humanitarian logistics management during natural disasters.

Keywords: Disaster resilience; Multimodal transport; Transport infrastructure; Humanitarian logistics; Emergency management; Multi-criteria decision making; FUCOM–MARCOS

1 Introduction

Crisis and humanitarian operations have become an increasingly important part of modern emergency management as natural disasters, climate-related hazards, technological accidents, and other disruptive events continue to threaten human life, property, infrastructure, and the normal functioning of society. The effectiveness of an emergency response depends substantially on whether logistics systems can deliver personnel, equipment, and humanitarian aid to affected areas within the required time. Transport is therefore a central component of humanitarian logistics because it directly shapes response time, resource availability, access to affected communities, and the continuity of emergency operations.

Transport planning in humanitarian settings differs markedly from routine logistics planning. Decisions must often be made under severe time pressure, with incomplete information, limited transport resources, and rapidly changing operating conditions. Floods, earthquakes, landslides, and large fires may damage roads and railways,

interrupt established routes, isolate settlements, and sharply reduce the capacity of the remaining network. A transport solution that is suitable under normal conditions may consequently become unavailable or unreliable during a disaster. Under these circumstances, the choice of transport solution is not simply an operational matter; it is also a question of infrastructure resilience and emergency management.

These challenges are particularly pronounced in Bosnia and Herzegovina because of its geographical, climatic, and infrastructural characteristics. Mountainous terrain, the uneven development of road and railway networks, capacity constraints on individual routes, and recurrent exposure to floods and landslides complicate the organisation of emergency transport. The floods and landslides that struck the Donja Jablanica area in October 2024 demonstrated that the success of humanitarian operations depends not only on the availability of vehicles, but also on the ability to select an appropriate transport arrangement within a short decision window. Such a selection must account simultaneously for delivery speed, network availability, reliability, safety, flexibility, and transport capacity.

Multimodal transport offers a possible response to these constraints. Combining road, rail, and air transport can reduce dependence on a single mode or route, provide alternatives when individual links are interrupted, and help maintain the movement of humanitarian supplies when part of the transport infrastructure is damaged or unavailable. It can also support more effective use of available logistics resources and strengthen the capacity of the transport system to adapt to disrupted operating conditions. However, multimodal arrangements introduce an additional layer of complexity. Each alternative performs differently across several quantitative and qualitative criteria: one may provide a shorter delivery time, another may involve lower costs, while others may offer greater capacity, reliability, safety, or flexibility. Selecting among these alternatives therefore requires a structured procedure capable of handling conflicting criteria.

Decision-making approaches based on a single indicator, or on a limited set of independently considered indicators, cannot adequately represent these trade-offs. Multi-Criteria Decision Making (MCDM) methods provide a systematic means of assessing several alternatives against multiple criteria expressed in different units and based on different types of information. Such methods have been widely applied in logistics, supply chain management, transport planning, supplier selection, risk management, and humanitarian logistics. Their relevance is especially clear in crisis operations, where decision-makers must compare technically different options despite limited time, constrained resources, and uncertainty about infrastructure conditions.

Among the available MCDM methods, the Full Consistency Method (FUCOM) and Measurement of Alternatives and Ranking according to Compromise Solution (MARCOS) offer complementary functions. FUCOM determines criterion weights through a limited number of pairwise comparisons while checking the consistency of expert judgements. This reduces the comparison burden and provides a structured representation of the relative importance assigned to the criteria. MARCOS ranks alternatives by examining their relationships with ideal and anti-ideal solutions. Its use allows transport alternatives with different performance profiles to be evaluated within a common decision framework. Integrating the two methods connects consistent criterion weighting with compromise-based ranking and is therefore suitable for transport decisions involving several competing operational requirements.

A substantial body of research has applied MCDM methods to logistics and transport problems. Much of this work, however, concerns supplier selection, logistics centre location, route optimisation, inventory management, and other decisions within commercial logistics systems. Fewer studies have examined the selection of multimodal transport arrangements for crisis and humanitarian operations, where damaged infrastructure, restricted access, limited transport resources, incomplete information, and short decision times alter the nature of the decision problem. Existing studies also provide limited guidance on how road, rail, air, and combined alternatives should be compared when the availability and reliability of transport infrastructure change during a disaster. This gap is particularly relevant to Bosnia and Herzegovina, where the physical configuration of the transport network and the country's exposure to floods and landslides require transport plans that remain workable under disrupted conditions.

This study addresses that problem by applying an integrated FUCOM–MARCOS model to the selection of a multimodal transport solution for humanitarian aid delivery under disaster-induced infrastructure disruptions. The model was examined through a logistics scenario based on the transport of 60 tonnes of humanitarian aid from the Logistics Base of the Armed Forces of Bosnia and Herzegovina at Rajlovac Barracks to the Donja Jablanica area following the floods and landslides of October 2024. The scenario provides a concrete setting in which road, rail, air, and combined transport alternatives can be assessed against the operational conditions encountered during a major natural disaster.

The study aims to identify the most suitable multimodal transport solution for crisis and humanitarian operations in Bosnia and Herzegovina. Seven criteria were included in the evaluation: delivery time, transport costs, transport network availability, reliability and resilience of the transport system, safety, flexibility, and transport capacity. FUCOM was used to determine the relative weights of these criteria, whereas MARCOS was applied to evaluate and rank five transport alternatives. A sensitivity analysis was subsequently conducted to examine whether changes in the criterion weights affected the resulting ranking.

The contribution of this study is threefold. First, it frames the selection of a humanitarian transport solution as a

resilience-oriented decision problem in which the availability and reliability of transport infrastructure are considered together with time, cost, safety, flexibility, and capacity. Second, it applies an integrated FUCOM–MARCOS procedure to compare single-mode and multimodal alternatives in a disaster setting, extending the use of these methods beyond conventional commercial logistics decisions. Third, it provides a transparent decision-support framework that can be used to examine transport options when individual network links are damaged, restricted, or unavailable. The framework is relevant to protection and rescue institutions, the Ministry of Defence of Bosnia and Herzegovina, the Armed Forces of Bosnia and Herzegovina, humanitarian organisations, and other bodies responsible for planning and implementing humanitarian aid transport during emergencies.

The remainder of the paper is organised as follows. Section 2 reviews research on humanitarian logistics, multimodal transport, and MCDM applications. Section 3 presents the theoretical foundations of FUCOM and MARCOS and explains their integration within the proposed decision model. Section 4 describes the case study, the evaluation criteria, and the transport alternatives. Section 5 reports and discusses the weighting and ranking results, while Section 6 presents the sensitivity analysis used to assess ranking stability. Section 7 concludes the paper by summarising the principal findings, acknowledging the research limitations, and identifying directions for further work.

2 Literature Review

Humanitarian logistics concerns the planning, organisation, and execution of aid flows during natural disasters, crises, and other emergencies. It differs from commercial logistics in both its priorities and its operating environment. Whereas commercial logistics is commonly driven by cost control and profitability, humanitarian logistics places greater emphasis on response time, delivery reliability, access to vulnerable populations, and the responsible use of scarce resources. Transport is consequently a core function of protection and rescue systems because delays or interruptions in the movement of supplies can directly affect the reach and effectiveness of a humanitarian response.

The literature consistently identifies logistics capacity as a determining factor in the delivery of equipment, food, water, medicine, and other essential supplies to affected areas. Van Wassenhove and Pedraza Martinez [1] describe logistics as a central element of humanitarian response, while Kovács and Spens [2] distinguish humanitarian supply chains by the uncertainty, information constraints, and time pressure under which they operate. Özdamar and Ertem [3] similarly observe that disaster-related transport decisions must often be made before decision-makers have complete information on infrastructure conditions, available capacities, and the needs of the affected population. Together, these studies establish that humanitarian transport planning requires decision procedures capable of operating with restricted information and within short response windows.

Research has subsequently broadened from the immediate movement of aid to the performance and organisational capacity of humanitarian supply chains. Patil et al. [4] examined the development of humanitarian supply chain performance measurement systems and identified strategies for improving their operation. In a related study, Patil et al. [5] investigated barriers to performance measurement and underlined the roles of coordination, information exchange, and resource availability in the management of humanitarian operations. Through a thematic review, Behl and Dutta [6] mapped the principal directions of humanitarian logistics research and called for more flexible and resilient supply chains supported by appropriate decision-making methods. This body of work indicates that operational performance cannot be separated from the institutional and informational conditions under which logistics decisions are made.

Digitalisation has emerged as another important strand of humanitarian logistics research. Dubey et al. [7] examined the relationships among big data analytics, organisational culture, trust, and collaborative performance in humanitarian supply chains, showing that analytical capacity must be accompanied by suitable organisational arrangements. Jayadi [8] considers the digitalisation of humanitarian supply chain performance management and draws attention to the role of data collection, processing, and exchange in time-sensitive decisions. Kurniawan [9] discusses digital innovation and systemic reform in developing countries, stressing that technological solutions must correspond to existing infrastructure and institutional capacity. Baharmand et al. [10] examine the application of blockchain in humanitarian supply chains, particularly its role in the transparency, traceability, and reliability of logistics processes. Although these studies differ in technological focus, they share the view that better information alone is insufficient unless it can be converted into timely and workable operational decisions.

This need to connect quantitative analysis with organisational action has also shaped broader interpretations of humanitarian logistics. Yáñez-Sandivari et al. [11] treat humanitarian logistics and emergency management as a sociotechnical and optimisation problem that requires quantitative methods to be integrated with the organisational dimensions of decision-making. Dohale et al. [12], drawing on the COVID-19 crisis, identify uncertainty, weak coordination, and limited resources as major barriers to the operationalisation of humanitarian supply chains. Their findings suggest that analytical models must account for the institutional and resource constraints that determine whether a proposed logistics solution can be implemented in practice.

More recent reviews have concentrated on innovation and unresolved research questions. Altay et al. [13] assessed

innovation in humanitarian logistics and supply chain management and identified the need for decision models suited to complex and changing crisis conditions. Köstepen and Selim [14], reviewing humanitarian logistics research on disaster management published between 2018 and 2024, highlighted logistics resilience, digitalisation, transport optimisation, and contemporary decision-support methods as prominent research directions. These findings show a gradual shift from static logistics planning towards approaches that can evaluate available resources and adjust transport and distribution decisions as operating conditions change.

The selection of a transport solution is therefore a central decision in crisis and humanitarian operations. Cost or delivery time alone cannot adequately represent the suitability of an alternative when infrastructure may be damaged and access to affected communities is uncertain. A defensible assessment must consider several interrelated and potentially conflicting criteria, including delivery time, transport cost, infrastructure availability, transport system reliability and resilience, safety, flexibility, capacity, and accessibility. The relative importance of these criteria may also differ from that observed in commercial transport. In a humanitarian emergency, for example, a higher-cost alternative may remain preferable if it provides faster or more reliable access to an isolated area. Such considerations are particularly relevant in countries with difficult terrain and unevenly developed transport networks, including Bosnia and Herzegovina.

Multimodal transport provides one means of reducing reliance on a single transport mode or network link. The combination of road, rail, air, and, where applicable, water transport can preserve alternative delivery paths when individual routes are interrupted. Rodrigue [15] explains that multimodal systems draw on the comparative operating characteristics of different transport modes. In a humanitarian context, this complementarity may also support continuity when part of the network is unavailable. Rodríguez-Espíndola et al. [16] incorporated intermodal transport and sustainability into a humanitarian logistics model, illustrating the relevance of combining transport modes in emergency resource planning. These studies provide a basis for treating multimodality not only as a question of transport efficiency, but also as a component of logistics and infrastructure resilience.

The selection of a transport mode or transport alternative under demanding operating conditions requires technical, economic, and operational considerations to be evaluated together. Starčević et al. [17] combined the Analytical Hierarchy Process (AHP) with Data Envelopment Analysis (DEA) to select off-road vehicles for transport operations. Their work showed how preference-based assessment and efficiency analysis can be combined when alternatives differ across several performance dimensions. Although the decision concerned transport equipment rather than humanitarian multimodality, the study is methodologically relevant because it addresses the selection of transport alternatives for a specific and restrictive operating environment.

Selecting a multimodal transport solution is consequently a multi-criteria decision problem. Road, rail, air, and combined alternatives may differ considerably in delivery time, cost, infrastructure requirements, reliability, resilience, safety, flexibility, and capacity. No alternative can therefore be judged solely by its performance against one criterion. MCDM methods provide a structured procedure for comparing these differences, assigning relative importance to the criteria, and producing a ranking that can be examined by decision-makers. Their role is particularly important when quantitative measurements and expert assessments must be considered within the same evaluation.

Within this methodological field, FUCOM and MARCOS have attracted attention because they address two distinct stages of an MCDM problem: criterion weighting and alternative ranking. Pamučar et al. [18] developed FUCOM to determine criterion weights with fewer comparisons than many conventional pairwise-comparison procedures while maintaining consistency among expert judgements. This feature is pertinent to emergency planning, where extensive elicitation may be impractical and inconsistent judgements may weaken the credibility of the resulting criterion weights.

FUCOM has since been used in several logistics and transport settings. Chakraborty and Saha [19] applied the method to group decision-making for forklift selection in a warehouse, demonstrating its use in logistics equipment assessment. Derse [20] integrated the Fuzzy Decision-Making Trial and Evaluation Laboratory (Fuzzy DEMATEL), FUCOM, and Step-wise Weight Assessment Ratio Analysis (SWARA) to prioritise measures for addressing barriers to green reverse logistics. These applications show that FUCOM can represent the relative importance of technical, operational, and managerial criteria across different logistics problems. They do not, however, directly address the weighting of criteria for multimodal humanitarian transport under disrupted infrastructure conditions.

MARCOS addresses the subsequent ranking stage by evaluating alternatives in relation to ideal and anti-ideal reference points. Stević et al. [21] introduced the method in a sustainable supplier-selection problem. Rong et al. [22] later developed a MARCOS approach based on cubic Fermatean fuzzy sets for the evaluation and selection of a cold-chain distribution centre, allowing uncertain and qualitative information to be incorporated into the assessment. Andrejić and Pajić [23] combined the Best-Worst Method, Quality Function Deployment, and MARCOS for strategic decision-making in cold-chain logistics, with MARCOS used to rank strategies and operational responses. Demir et al. [24] reviewed the development and application of MARCOS between 2020 and 2024 and documented its growing use across multiple decision fields. Collectively, these studies demonstrate the method's capacity to distinguish among logistics alternatives with different performance profiles, although most applications remain outside disaster-related

transport planning.

The complementary characteristics of FUCOM and MARCOS are relevant to crisis operations, where experts may need to specify priorities and compare transport options within a limited period and with incomplete information. Their integration remains a comparatively recent development in MCDM research. Stević and Brković [25] combined FUCOM and MARCOS to evaluate human resources in a transport company, demonstrating the feasibility of linking FUCOM-based weighting with MARCOS-based ranking in the transport sector. Blagojević et al. [26] subsequently applied the two methods to logistics service-provider selection in Bosnia and Herzegovina. FUCOM was used to determine criterion weights, MARCOS was used to rank the providers, and sensitivity analysis was employed to examine ranking stability. These studies confirm the operational compatibility of the two methods, but their decision settings differ from the selection of transport arrangements during a natural disaster.

This literature shows that combining weighting, efficiency, and ranking procedures can make the basis of a decision more explicit and allow the stability of the outcome to be examined. Nevertheless, methodological sophistication alone does not resolve the substantive problem of how to compare single-mode and multimodal transport alternatives when infrastructure availability changes during an emergency.

Despite the breadth of MCDM applications in logistics and transport, comparatively little research has addressed the integrated selection of multimodal transport solutions for crisis and humanitarian operations. Existing studies have concentrated primarily on humanitarian supply chain performance, the location of humanitarian facilities, aid distribution, supplier selection, inventory optimisation, or the assessment of individual transport and logistics options. These contributions address important elements of humanitarian operations but provide limited guidance on choosing among road, rail, air, and combined alternatives when parts of the transport network are degraded or unavailable. The use of an integrated FUCOM–MARCOS model for crisis and humanitarian transport planning in Bosnia and Herzegovina also remains largely unexplored.

The resulting research gap has both substantive and methodological dimensions. Substantively, the selection of a multimodal alternative under partial infrastructure disruption has received less attention than facility location, distribution, and supply chain performance. Methodologically, previous studies establish the usefulness of resilience assessment, digital information, flexible transport arrangements, and MCDM procedures, but they do not fully connect these elements within a single transport-selection framework. In particular, the selection of a road, rail, air, or integrated alternative under damaged or capacity-constrained infrastructure remains insufficiently examined in the context of Bosnia and Herzegovina.

This study responds to that gap through an integrated FUCOM–MARCOS model for evaluating multimodal transport alternatives under uncertainty, time pressure, and restricted infrastructure availability. FUCOM was used to determine the relative importance of the evaluation criteria, while MARCOS was used to compare and rank the transport alternatives. By applying this framework to humanitarian aid delivery during the October 2024 floods and landslides, the study extends integrated MCDM analysis to a resilience-oriented transport decision in Bosnia and Herzegovina. The resulting model provides a structured basis for institutions responsible for planning and implementing crisis and humanitarian operations to examine transport alternatives when individual network components are damaged, constrained, or unavailable.

3 Methods

The selection of a multimodal transport solution under disaster-induced infrastructure disruptions requires the simultaneous assessment of operational, economic, safety, and resilience-related criteria. To structure this decision problem, an integrated FUCOM–MARCOS model was applied. FUCOM was used to determine the relative weights of the evaluation criteria from consistent expert judgements, while MARCOS was employed to assess and rank the transport alternatives according to their relationships with the ideal and anti-ideal solutions. The integrated procedure provides a transparent basis for identifying the most suitable transport arrangement when network availability, transport capacity, and operating conditions are constrained by a natural disaster.

3.1 Full Consistency Method (FUCOM)

The FUCOM was developed by Pamučar et al. [18] for determining the weights of criteria. The procedure for determining the criterion weights using FUCOM is presented below.

Step 1: Criteria from the predefined set $\{C_1, C_2, \dots, C_n\}$ are ranked according to their significance in the following way:

$$C_{j(1)} > C_{j(2)} > \dots > C_{j(n)} \quad (1)$$

where, $C_{j(r)}$ denotes the criterion occupying the r -th position in the ranking, j denotes the criterion index, r denotes the ranking position, and n is the total number of criteria.

Step 2: The ranked criteria are compared in pairs, and the comparative priority $\varphi_{k/(k+1)}$ between two consecutively ranked criteria is determined, where $k = 1, 2, \dots, n - 1$.

$$\Phi = (\varphi_{1/2}, \varphi_{2/3}, \dots, \varphi_{k/(k+1)}) \quad (2)$$

where, Φ denotes the vector of comparative priorities, and $\varphi_{k/(k+1)}$ represents the comparative priority of criterion $C_{j(k)}$ relative to criterion $C_{j(k+1)}$.

Step 3: The final weighting coefficients w_j are calculated so that they satisfy two conditions:

First, the ratio of weighting coefficients is equal to the comparative priority between the observed criteria $\varphi_{k/(k+1)}$ given in Eq. (2); that is, the following condition is satisfied:

$$\frac{w_{j(k)}}{w_{j(k+1)}} = \varphi_{k/(k+1)} \quad (3)$$

where, $w_{j(k)}$ and $w_{j(k+1)}$ denote the weighting coefficients of the criteria ranked k -th and $(k + 1)$ -th, respectively.

Second, in addition to the condition expressed in Eq. (3), the final weighting coefficients should satisfy the condition of mathematical transitivity:

$$\varphi_{k/(k+1)} \times \varphi_{(k+1)/(k+2)} = \varphi_{k/(k+2)}.$$

Accordingly, the following additional condition should be satisfied:

$$\frac{w_{j(k)}}{w_{j(k+2)}} = \varphi_{k/(k+1)} \times \varphi_{(k+1)/(k+2)} \quad (4)$$

where, $w_{j(k+2)}$ denotes the weighting coefficient of the criterion ranked $(k + 2)$ -th, and $\varphi_{(k+1)/(k+2)}$ denotes the comparative priority of criterion $C_{j(k+1)}$ relative to criterion $C_{j(k+2)}$.

Based on these two conditions, the final model for determining the criteria weights is defined, as given in Eq. (5):

$$\begin{aligned} \min \quad & \chi \\ \text{s.t.} \quad & \left| \frac{w_{j(k)}}{w_{j(k+1)}} - \varphi_{k/(k+1)} \right| \leq \chi, \quad \forall j, \\ & \left| \frac{w_{j(k)}}{w_{j(k+2)}} - \varphi_{k/(k+1)} \varphi_{(k+1)/(k+2)} \right| \leq \chi, \quad \forall j, \\ & \sum_{j=1}^n w_j = 1, \\ & w_j \geq 0, \quad \forall j. \end{aligned} \quad (5)$$

where, χ denotes the deviation from full consistency. By solving Eq. (5), the final weight vector $\mathbf{w} = (w_1, w_2, \dots, w_n)^T$ and the corresponding deviation from full consistency χ are obtained [18], with T denoting the transpose.

3.2 Measurement of Alternatives and Ranking According to Compromise Solution (MARCOS) Method

The MARCOS method is performed through the following steps [21]:

Step 1: The decision matrix contains m alternatives evaluated according to n criteria.

Step 2: The initial matrix is extended by adding the ideal alternative (AI) and anti-ideal alternative (AAI), as shown in Eq. (6):

$$X = \begin{matrix} & \begin{matrix} C_1 & C_2 & \cdots & C_n \end{matrix} \\ \begin{matrix} \text{AAI} \\ A_1 \\ A_2 \\ \vdots \\ A_m \\ \text{AI} \end{matrix} & \begin{bmatrix} x_{\text{AAI},1} & x_{\text{AAI},2} & \cdots & x_{\text{AAI},n} \\ x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \\ x_{\text{AI},1} & x_{\text{AI},2} & \cdots & x_{\text{AI},n} \end{bmatrix} \end{matrix} \quad (6)$$

where, X denotes the extended decision matrix, A_i denotes the i -th alternative, C_j denotes the j -th criterion, and x_{ij} represents the performance value of alternative A_i with respect to criterion C_j .

Depending on whether a criterion belongs to the benefit (B) or cost (C) group, AAI and AI are calculated using Eqs. (7) and (8):

$$\text{AAI} = \begin{cases} \min_i x_{ij}, & j \in B, \\ \max_i x_{ij}, & j \in C. \end{cases} \quad (7)$$

$$AI = \begin{cases} \max_i x_{ij}, & j \in B, \\ \min_i x_{ij}, & j \in C. \end{cases} \quad (8)$$

Step 3: The extended matrix X is normalised using Eqs. (9) and (10):

$$n_{ij} = \frac{x_{AI,j}}{x_{ij}}, \quad j \in C \quad (9)$$

$$n_{ij} = \frac{x_{ij}}{x_{AI,j}}, \quad j \in B \quad (10)$$

where, n_{ij} denotes the normalised performance value of alternative A_i with respect to criterion C_j , and $x_{AI,j}$ denotes the ideal value of criterion C_j .

Step 4: The normalised matrix N is multiplied by the corresponding criterion weights to obtain the weighted normalised matrix V , as shown in Eq. (11).

$$V_{ij} = n_{ij} \times w_j \quad (11)$$

where, V_{ij} denotes the weighted normalised performance value of alternative A_i with respect to criterion C_j .

Step 5: The utility degree of each alternative, K_i , is calculated relative to both AAI and AI using Eqs. (12) and (13):

$$K_i^- = \frac{S_i}{S_{AAI}} \quad (12)$$

$$K_i^+ = \frac{S_i}{S_{AI}} \quad (13)$$

where, K_i^- and K_i^+ denote the utility degrees of alternative A_i relative to AAI and AI, respectively, while S_{AAI} and S_{AI} denote the sums of the weighted normalised values for AAI and AI, respectively. The value of S_i is calculated using Eq. (14):

$$S_i = \sum_{j=1}^n V_{ij} \quad (14)$$

Step 6: The final utility function $f(K_i)$ is calculated based on the position of alternative A_i relative to AAI and AI, as shown in Eq. (15).

$$f(K_i) = \frac{K_i^+ + K_i^-}{1 + \frac{1 - f(K_i^+)}{f(K_i^+)} + \frac{1 - f(K_i^-)}{f(K_i^-)}} \quad (15)$$

The auxiliary utility functions $f(K_i^-)$ and $f(K_i^+)$, which represent the utilities relative to AAI and AI, respectively, are calculated using Eqs. (16) and (17):

$$f(K_i^-) = \frac{K_i^+}{K_i^+ + K_i^-} \quad (16)$$

$$f(K_i^+) = \frac{K_i^-}{K_i^+ + K_i^-} \quad (17)$$

Step 7: The alternatives are ranked according to their final utility function values.

4 Case Study

For the purpose of verifying the proposed model for selecting a multimodal transport solution, the Ministry of Defence of Bosnia and Herzegovina was selected as a representative example of an institution possessing the organisational, logistical, and transport capacities required for the implementation of crisis and humanitarian operations. The selection of this institution is based on its legal responsibility to provide, through the Armed Forces of Bosnia and Herzegovina, support to civilian authorities in cases of natural and other disasters, as well as on the fact that it possesses various types of transport resources enabling the implementation of multimodal transport.

In accordance with the Law on Defence of Bosnia and Herzegovina, the Ministry of Defence of Bosnia and Herzegovina is responsible for planning, organising, and managing the defence system, developing and maintaining the operational capabilities of the Armed Forces of Bosnia and Herzegovina, managing human and material resources, and coordinating activities related to providing support to civilian institutions in emergency situations. In crisis situations, such as floods, landslides, earthquakes, large fires or other natural and technical-technological disasters,

the Armed Forces of Bosnia and Herzegovina are engaged in conducting search and rescue operations, evacuating the population, distributing humanitarian aid, establishing communication and transport links, and providing engineering and logistical support to affected areas.

From a logistics support perspective, the Ministry of Defence of Bosnia and Herzegovina has considerable transport capacities, including road vehicles with different load capacities, specialised military vehicles designed for difficult terrain, aviation assets for the rapid transport of personnel and material resources, and engineering equipment used to establish temporary routes and overcome infrastructure obstacles. The integration of these resources enables the development of different multimodal transport scenarios, thereby improving the flexibility and resilience of the logistics system, particularly when parts of the transport infrastructure are damaged or completely unavailable.

A particular advantage of the system is its high level of organisation, clearly defined hierarchical command structure, and the ability to rapidly mobilise human and material resources. Such an organisational structure enables efficient planning and coordination of transport activities, as well as rapid adaptation to changes occurring during crisis and humanitarian operations. At the same time, decision-making under such conditions is further complicated by limited information availability, dynamic changes in field conditions, transport infrastructure constraints, budgetary limitations, and the need to coordinate a larger number of institutions involved in the emergency management system.

Due to the aforementioned characteristics, the Ministry of Defence of Bosnia and Herzegovina represents a suitable environment for the application of an MCDM model. Within this case study, alternative multimodal transport options that may be employed during crisis and humanitarian operations were considered, with delivery time, transport cost, transport infrastructure availability, reliability and resilience of the transport system, safety, flexibility, and transport capacity used as evaluation criteria. By applying the integrated FUCOM–MARCOS model, the criterion weights were determined and the considered alternative transport solutions were ranked, with the aim of identifying the optimal multimodal transport model for the conditions in Bosnia and Herzegovina.

The proposed case study does not only demonstrate the applicability of the developed model but also provides a basis for improving the planning process of logistics support in emergency situations. The obtained results can serve as decision support for decision-makers within the Ministry of Defence of Bosnia and Herzegovina, the Armed Forces of Bosnia and Herzegovina, as well as other institutions responsible for emergency management during the planning and implementation of crisis and humanitarian operations.

During the first days of October 2024, Bosnia and Herzegovina was hit by heavy rainfall that triggered severe floods and landslides in the Herzegovina-Neretva Canton. The most serious damage occurred in the Donja Jablanica area, where large landslides destroyed residential buildings, damaged road and railway connections, and left some settlements completely cut off from the rest of the country. Due to the scale of the resulting damage, numerous domestic and international rescue services were engaged, including the Ministry of Defence of Bosnia and Herzegovina and the Armed Forces of Bosnia and Herzegovina, which participated in search and rescue operations, population evacuation, humanitarian aid delivery, and the establishment of logistics support for affected areas.

For the purpose of verifying the proposed FUCOM–MARCOS model, a crisis scenario based on the aforementioned event was analysed. For the purpose of this case study, a logistics scenario was defined based on the engagement of the Ministry of Defence of Bosnia and Herzegovina and the Armed Forces of Bosnia and Herzegovina during the floods and landslides that affected the Donja Jablanica area in early October 2024. Within the defined scenario, the organisation of the transport of 60 tonnes of humanitarian aid from the Logistics Base of the Armed Forces of Bosnia and Herzegovina at Rajlovac Barracks to the Donja Jablanica area was analysed. The humanitarian cargo included drinking water, food packages, medicines and medical supplies, electrical generators, water pumps, tents, blankets, and other equipment necessary for providing assistance to the affected population.

At the time of transport implementation, the road infrastructure towards Jablanica was partially damaged due to landslides and floods, while the railway line was out of operation on certain sections due to damage to the tracks and embankments. At the same time, air transport was limited by available helicopter capacities, meteorological conditions, and the possibility of organising landing sites in the affected area. Under such circumstances, it was necessary to consider several possible transport technologies that would enable the timely and efficient delivery of humanitarian aid.

Based on the available resources of the Ministry of Defence of Bosnia and Herzegovina and the Armed Forces of Bosnia and Herzegovina, transport alternatives were defined as the subject of multi-criteria evaluation. For each alternative, the delivery time, total transport costs, transport network availability, reliability and resilience of the transport system, safety, flexibility, and transport capacity were analysed. The quantitative data used in the research represent the results of a logistics scenario simulation based on the actual characteristics of transport resources and infrastructure, while qualitative criteria were evaluated by experts in the fields of logistics, transport, and crisis management.

The proposed scenario represents a realistic example of a complex logistics problem in which it is necessary to select an optimal transport alternative for humanitarian aid delivery within a very short period of time, under conditions of partially degraded transport infrastructure and a high level of uncertainty. Therefore, it is suitable for

the application of the integrated FUCOM–MARCOS model as a decision-support tool in crisis and humanitarian operations.

4.1 Definition of Possible Transport Alternatives

One of the key steps in developing a model for selecting an optimal multimodal transport solution is the identification of transport alternatives that can be applied in crisis and humanitarian operations. The selection of an appropriate transport alternative under such conditions represents a complex decision-making problem, as it depends on multiple interrelated criteria, including delivery time, transport network availability, transport costs, reliability and resilience of the transport system, safety, flexibility, and transport capacity.

Unlike conventional logistics systems, transport in crisis and humanitarian operations takes place under conditions of high uncertainty, limited information, and frequently damaged or partially functional infrastructure. Floods, landslides, earthquakes, snow blockages, and other emergency events may significantly restrict the use of certain transport modes, making it necessary to consider various modal and multimodal options that ensure the continuity of humanitarian aid and logistics resource delivery.

Based on the available transport capacities of the Ministry of Defence of Bosnia and Herzegovina, the characteristics of the transport infrastructure of Bosnia and Herzegovina, and the requirements of crisis and humanitarian operations, this research defined five alternative transport solutions that are the subject of multi-criteria evaluation using the integrated FUCOM–MARCOS model.

A1: Direct road transport from Rajlovac Barracks to Donja Jablanica. This transport alternative involves the direct transport of humanitarian aid from the logistics base of the Armed Forces of Bosnia and Herzegovina in the Rajlovac Barracks to the Donja Jablanica area using military cargo vehicles. Road transport enables the direct delivery of aid to the final destination without the need for cargo transshipment, which contributes to a simpler organisation of the transport process and a high degree of operational flexibility. During the crisis events in October 2024, road transport was the main method of delivering humanitarian aid, but its efficiency was significantly affected by the state of the road infrastructure. Floods, landslides, and road damage on certain sections of the main road network caused transport delays, necessitated the use of alternative routes, and increased the time required for transport operations. Despite the aforementioned limitations, direct road transport enables the transport of larger quantities of humanitarian aid and the simple engagement of available transport capacities, which is why it is one of the main transport alternatives in crisis and humanitarian operations.

A2: Rail transport to the nearest available logistics point, followed by road distribution via alternative routes. This transport alternative involves delivering humanitarian aid by rail from the logistics centre to the nearest railway terminal with operational infrastructure, where the cargo is then transferred to road vehicles for further distribution to the Donja Jablanica area. The application of this transport alternative enables the efficient transportation of larger quantities of humanitarian aid over medium and long distances, with lower transport costs and higher transport capacity compared with road transport alone. However, during the floods and landslides in October 2024, the railway infrastructure on sections of the railway line towards Jablanica was damaged, preventing the implementation of complete rail transport to the final destination. Therefore, it was necessary to transfer humanitarian aid to road vehicles and organise the final distribution to the affected area. Although additional transshipment increases the total transport time, this transport alternative enables the efficient utilisation of the high capacity of rail transport while maintaining the flexibility of road transport in the final stage of the logistics chain.

A3: Direct air transport by helicopters of the Armed Forces of Bosnia and Herzegovina. This transport alternative involves the direct delivery of humanitarian aid by helicopters from the Logistics Base at Rajlovac Barracks to the Donja Jablanica area. Air transport is primarily used for the rapid delivery of aid when road and rail connections are completely or partially disrupted, or when ground transport cannot be carried out within an acceptable time frame. During the floods and landslides in October 2024, helicopters of the Armed Forces of Bosnia and Herzegovina were used to evacuate the population, transport rescue teams, and deliver humanitarian aid to isolated areas, demonstrating their important role in crisis response and humanitarian operations. The main advantage of this transport alternative is the short transport time and the ability to reach areas that are inaccessible by land. However, its use is limited by the relatively small amount of cargo that can be transported per flight, high operating costs, weather conditions, aircraft availability, and the possibility of safe landing or cargo unloading in the affected area. Despite these limitations, direct air transport can be considered the most effective option for the rapid delivery of critical humanitarian aid during the initial phase of a crisis.

A4: Combined road–air transport (truck + helicopter). This transport alternative combines road and air transport. Humanitarian aid is first transported by truck from the Logistics Base of the Armed Forces of Bosnia and Herzegovina at Rajlovac Barracks to the nearest safe location suitable for organising air transport. The cargo is then transferred to helicopters and delivered to the Donja Jablanica area. Such a transport organisation enables the rational use of road transport on sections with preserved road infrastructure, while air transport is used to overcome sections where roads are disrupted or inaccessible due to landslides, floods, or other consequences of natural disasters.

During the floods and landslides in October 2024, this type of transport organisation represented one of the most efficient solutions for delivering humanitarian aid to isolated areas, as it enabled the rapid deployment of helicopter capacities while simultaneously reducing their engagement time on longer transport routes. Compared with exclusive air transport, this alternative enables more efficient utilisation of available aviation resources and reduces overall transport costs, while compared with exclusive road transport, it provides significantly shorter delivery times and greater accessibility to difficult-to-reach locations. The disadvantage of this transport alternative lies in the need to organise cargo transshipment, coordinate two transport technologies, and the dependence of the air transport segment on meteorological conditions and the availability of helicopter capacities.

A5: Integrated multimodal transport (road–rail–air). This transport alternative involves the integrated application of road, rail, and air transport in accordance with transport infrastructure availability and operational conditions in the field. Humanitarian aid is initially transported by rail or road to the nearest functional logistics point, from where it is further distributed, depending on the infrastructure conditions and terrain characteristics, by road transport resources or helicopters of the Armed Forces of Bosnia and Herzegovina to the final destination in Donja Jablanica. Such a transport organisation enables the optimal utilisation of the advantages of each transport mode, where rail transport provides high transport capacity on main routes, road transport ensures flexible distribution along accessible sections, and air transport enables access to areas that have become completely or partially isolated from land connections due to floods and landslides. During the floods and landslides in October 2024, the integrated multimodal approach would represent the most flexible logistics solution, as it enables the adaptation of the transport process to changes in infrastructure conditions and the availability of transport resources. At the same time, this transport alternative requires a high level of coordination among different participants in the transport chain, the organisation of multiple transshipment operations, and efficient management of logistics resources, which increases the complexity of transport planning and implementation. Despite these challenges, integrated multimodal transport enables a high level of reliability and resilience of the transport system, efficient utilisation of available transport capacities, and continuity of humanitarian aid delivery under conditions of severely disrupted transport infrastructure.

The defined transport alternatives represent realistic logistics scenarios that can be applied within the crisis and humanitarian response system of Bosnia and Herzegovina. Their performances differ according to delivery time, transport cost, transport network availability, reliability and resilience of the transport system, safety, flexibility, and transport capacity, which makes their objective evaluation possible only through the application of an MCDM model. In the following sections of the paper, these alternatives will be evaluated using the integrated FUCOM–MARCOS model, with the aim of identifying the optimal multimodal transport solution for crisis and humanitarian operations in Bosnia and Herzegovina.

4.2 Definition of Criteria

The definition of appropriate criteria represents one of the most important phases in the development of an MCDM model, as the quality of the final decision largely depends on the selection of criteria that reliably describe the performance of the considered alternatives. In problems related to the selection of transport solutions for crisis and humanitarian operations, criteria must enable a comprehensive assessment of the operational, economic, infrastructural, and safety aspects of the transport system, while simultaneously considering the specific characteristics of operating under conditions of high uncertainty and time pressure.

Unlike commercial transport systems, where the primary objectives are cost optimisation and increased profitability, humanitarian operations prioritise response speed, accessibility to affected areas, delivery reliability, and the ability of the transport system to operate under conditions of damaged or partially unavailable infrastructure. Therefore, the criteria used in this study encompass the technical, operational, economic, and organisational characteristics of different transport alternatives.

The criteria were defined based on a detailed review of relevant scientific literature covering humanitarian logistics, multimodal transport, crisis management, and MCDM, as well as consultations with experts in logistics, transport, and crisis management. Particular attention was paid to selecting criteria that are mutually independent, representative of the main characteristics of the transport alternatives, and applicable to the specific conditions in Bosnia and Herzegovina.

Based on the analysis, seven criteria were identified as key performance indicators for multimodal transport solutions in crisis and humanitarian operations (Table 1). These criteria cover both quantitative and qualitative aspects of the transport system. Quantitative criteria are expressed using numerical values, allowing for a more objective comparison of the available alternatives, while qualitative criteria describe factors that cannot be measured directly but can still have a significant influence on decision-making. Considering both types of criteria provides a more comprehensive and realistic evaluation of the performance of the proposed transport alternatives.

C1: Delivery time. Delivery time is one of the most important operational criteria in crisis and humanitarian operations because it directly affects the speed of response and the timely provision of assistance to the affected population. In natural disasters and other emergency situations, even relatively short delays can have serious

consequences for human safety and the overall effectiveness of humanitarian operations. This criterion includes the total time required for the implementation of the transport process, including transport preparation time, loading, transshipment between different transport modes, transportation itself, and unloading at the final destination. Since the objective is to achieve the shortest possible transport time, this criterion is considered a cost-type (min) criterion.

Table 1. Classification of criteria used in the model

	Criteria	Type of Criterion	Optimisation Objective	Evaluation Method/Unit
C1	Delivery time	Quantitative	Min	Hours (h)
C2	Transport cost	Quantitative	Min	BAM or EUR
C3	Transport network availability	Qualitative	Max	Expert assessment
C4	Reliability and resilience of the transport system	Qualitative	Max	Expert assessment
C5	Safety	Qualitative	Max	Expert assessment
C6	Flexibility	Qualitative	Max	Expert assessment
C7	Transport capacity	Quantitative	Max	Tonnes (t), m ³ or number of vehicles

Note: BAM = Bosnia and Herzegovina Convertible Mark; EUR = European Monetary Unit.

C2: Transport cost. Transport cost represents the basic economic indicator of transport system efficiency and includes all direct and indirect costs incurred during the implementation of the transport process. This group includes the costs of using transport resources, fuel, human resources, maintenance, transshipment, and other logistics activities associated with transport implementation. Although response speed has priority over the economic aspect in humanitarian operations, the rational utilisation of available financial resources represents an important element of system sustainability, particularly in situations where multiple humanitarian interventions are conducted simultaneously. This criterion also belongs to the group of cost-type (min) criteria.

C3: Transport network availability. Transport network availability represents the degree of availability, functionality, and accessibility of transport infrastructure required for the implementation of a specific transport alternative. This criterion includes the condition of road, rail, and air connections, the possibility of accessing affected areas, and the ability of the transport system to operate despite infrastructure damage caused by natural disasters or other emergency events. A higher level of transport network availability enables more reliable and efficient implementation of humanitarian operations; therefore, this criterion belongs to the group of benefit-type (max) criteria.

C4: Reliability and resilience of the transport system. Reliability and resilience of the transport system represent the ability of a transport alternative to ensure the continuous, timely, and safe implementation of the transport process, as well as to maintain or restore its functionality within the shortest possible time under conditions of disruptions caused by natural disasters, infrastructure damage, adverse weather conditions, or other emergency circumstances characteristic of crisis and humanitarian operations. This criterion includes the probability that transport will be completed within the planned time frame while maintaining the planned quantity, quality, and integrity of the cargo, as well as the ability of the transport system to effectively adapt to disruptions, utilise alternative transport routes, and maintain the continuity of humanitarian aid delivery. Reliability and resilience depend on the stability and availability of transport infrastructure, the organisation of the logistics system, coordination among participants in the transport chain, availability of alternative transport resources, and the efficiency of crisis management. Transport alternatives achieving a higher level of reliability and resilience provide greater security in the implementation of logistics activities, reduce the risk of transport flow disruptions, and contribute to a more effective response to crisis and humanitarian events. Since a higher level of reliability and resilience represents a desirable characteristic of a transport system, this criterion is classified as a benefit-type (max) criterion.

C5: Safety. Safety represents one of the key criteria in the selection of a multimodal transport solution for crisis and humanitarian operations, as it directly affects the protection of human lives, preservation of humanitarian aid, and uninterrupted execution of the transport process. Under conditions of natural disasters and other emergency events, the transport system is exposed to numerous risks, including damage to transport infrastructure, adverse weather conditions, traffic accidents, technical failures of transport resources, as well as security threats that may jeopardise the implementation of logistics activities. In this study, safety encompasses the level of protection of personnel, transport resources, and humanitarian cargo during all phases of the transport process, including loading, transportation, transshipment, and unloading. Particular attention is given to the ability of the transport system to minimise the risk of aid damage or loss, as well as to ensure the safe engagement of personnel involved in the implementation of crisis and humanitarian operations. Transport alternatives that provide a higher level of operational safety contribute to greater

reliability of the overall logistics system and reduce the probability of undesirable events. Since a higher level of safety represents a desirable characteristic of the transport system, this criterion is classified as a benefit-type (max) criterion.

C6: Flexibility. Flexibility represents the ability of the transport system to rapidly adapt to changes occurring during the implementation of crisis and humanitarian operations. Unlike standard logistics systems, where transport flows are predefined and relatively stable, crisis situations are characterised by uncertainty, frequent changes in priorities, limited availability of transport networks, and the need to make decisions within a very short time frame. Therefore, flexibility represents one of the most important characteristics of a multimodal transport system. In this study, flexibility refers to the ability of a transport alternative to enable changes in transport routes, the combination of different transport modes, adaptation to changes in cargo quantity and type, as well as the rapid mobilisation of available transport capacities in accordance with the development of the crisis situation. Transport systems with a high level of flexibility enable more efficient management of logistics flows and greater resilience to disruptions caused by natural disasters or other emergency events. Therefore, flexibility is classified as a benefit-type (max) criterion.

C7: Transport capacity. Transport system capacity represents the maximum quantity of people, equipment, or humanitarian aid that can be transported within a specific time period using available transport resources. In humanitarian operations, this criterion has particular importance, as the volume of required assistance often significantly exceeds available logistics capacities, especially during major natural disasters and other emergency situations. Transport capacity depends on the carrying capacity of transport resources, the availability of logistics resources, the possibility of organising multiple transport cycles, and the level of utilisation of existing transport infrastructure. Transport alternatives with higher capacity enable more efficient distribution of humanitarian aid, reduce the number of individual transport operations, and ensure more rational utilisation of available resources. At the same time, higher transport capacity contributes to the faster restoration of the normal functioning of the logistics system following the occurrence of a crisis event. For these reasons, capacity is classified as a benefit-type (max) criterion.

Based on the previously defined criteria, a set of indicators was established to enable a comprehensive assessment of the performance of the considered multimodal transport alternatives. The selected criteria encompass operational, economic, infrastructural, and organisational characteristics of the transport system, thereby ensuring an objective and systematic evaluation of alternatives in accordance with the requirements of crisis and humanitarian operations. The defined set of criteria represents the basis for determining their relative weights using FUCOM, while in the subsequent phase of the research it will be used for ranking multimodal transport alternatives through the application of the MARCOS method.

5 Research Results

The results of the application of the integrated FUCOM–MARCOS model are presented through the process of determining the weighting coefficients of the criteria, forming the decision matrix, evaluating the defined multimodal transport alternatives, and ranking them using the MARCOS method. In addition, a sensitivity analysis was conducted to examine the stability and reliability of the obtained results under changes in the relative importance of the criteria.

5.1 Research Results Obtained Using FUCOM

By applying FUCOM, experts first ranked the defined criteria according to their relative importance, after which comparative priorities between consecutively ranked criteria were determined. Based on expert assessments and the mathematical model of FUCOM, normalised weighting coefficients were calculated for each criterion (Table 2). The obtained weights represent the relative importance of individual criteria in the process of selecting the optimal multimodal transport solution for crisis and humanitarian operations in Bosnia and Herzegovina and provide the basis for further evaluation and ranking of transport alternatives using the MARCOS method.

The results presented in Table 2 show a high level of agreement among the five decision-makers, indicating that the expert assessments are consistent and that FUCOM is suitable for determining the relative importance of the criteria. The average weighting coefficients show that delivery time (C1) is the most important criterion, with a weight of 0.236. This result is expected, as the timely delivery of humanitarian aid is a key requirement for an effective response in crisis and humanitarian operations.

The next most important factor is the reliability and resilience of the transport system (C4) with a weight of 0.184, which reflects the ability of the transport system to ensure continuous and timely implementation of transport activities even in conditions of disruption of the transport infrastructure. Safety (C5) ranks third with a weight of 0.154, which confirms the importance of protecting people, transport vehicles and humanitarian cargo during the implementation of the transport process.

The transport network availability (C3) with a weight of 0.132 is also of relatively high importance, since the functionality of road, rail and air infrastructure directly affects the possibility of implementing certain multimodal

transport solutions. Flexibility (C6), with a weight of 0.119, is an important criterion due to the need to quickly adapt to changes in transport routes, available resources and operational conditions that characterize crisis situations.

Table 2. Weighting coefficient values

	Criteria	DM1	DM2	DM3	DM4	DM5	w_j
C1	Delivery time	0.238	0.235	0.237	0.236	0.234	0.236
C2	Transport cost	0.062	0.064	0.063	0.063	0.063	0.063
C3	Transport network availability	0.131	0.133	0.132	0.132	0.132	0.132
C4	Reliability and resilience of the transport system	0.185	0.183	0.184	0.184	0.184	0.184
C5	Safety	0.154	0.153	0.155	0.154	0.154	0.154
C6	Flexibility	0.118	0.120	0.119	0.119	0.119	0.119
C7	Transport capacity	0.112	0.111	0.112	0.112	0.112	0.112

Note: DM = decision maker; DM1–DM5 = the five experts participating in the criterion weighting process; C_j = the j -th evaluation criterion ($j = 1, \dots, 7$); w_j = final aggregated normalised weighting coefficient of criterion C_j .

Transport capacity (C7) was given a weight of 0.112, indicating that the ability to transport larger quantities of humanitarian aid is a significant, but not a decisive factor in choosing a transport alternative. The lowest weight is given to transport cost (C2) with a value of 0.063, confirming that in crisis and humanitarian operations the economic aspect is not the primary decision-making criterion. In such situations, priority is given to the speed of response, reliability and safety of transport implementation, while cost optimisation is of secondary importance.

The resulting distribution of weight coefficients is fully aligned with the specifics of humanitarian logistics and represents a realistic basis for further multi-criteria evaluation and ranking of defined multimodal transport alternatives using the MARCOS method.

5.2 Results of the MARCOS Method Application

After determining the relative importance of the criteria using FUCOM, the defined multimodal transport alternatives were evaluated and ranked using the MARCOS method. The aim of this phase was to determine which transport alternative performs best according to the selected criteria and can therefore be considered the most suitable solution for crisis and humanitarian operations in Bosnia and Herzegovina.

The performance of the transport alternatives was assessed by five experts with experience in logistics, transport, and crisis management. A seven-point Likert scale was used to evaluate the qualitative criteria, and the average scores were then used in the subsequent calculations. Quantitative criteria were determined based on the available technical and operational characteristics of each transport alternative. This approach allowed objective indicators and expert evaluations to be combined into a single decision matrix.

To ensure a more objective evaluation, the transport alternatives were assigned codes (A1–A5) without revealing any information that could indicate a preferred option during the assessment. This helped reduce the potential influence of subjective expert opinions and contributed to greater objectivity in the final results.

In accordance with the MARCOS methodology, an extended decision matrix was first constructed, including the considered transport alternatives as well as AI and AAI. This matrix serves as the basis for the subsequent normalisation of the data, calculation of the weighted matrix, and determination of the utility functions used to obtain the final ranking of the multimodal transport alternatives. Table 3 presents the extended initial decision matrix prepared according to the second step of the MARCOS method, as defined by Eqs. (6)–(8).

After forming the extended initial decision matrix, the next step of the MARCOS method involves its normalisation using Eqs. (9) and (10). The normalisation procedure enables the transformation of the values of all criteria into a comparable scale, whereby different expressions are applied to cost-type (min) and benefit-type (max) criteria.

Examples of the normalisation calculations for alternative A1 (direct road transport) are provided below. For the cost-type criterion C1 (delivery time): $n_{11} = \frac{x_{A1,1}}{x_{11}} = \frac{4.0}{9.5} = 0.4211$. For the benefit-type criterion C3 (transport network availability): $n_{13} = \frac{x_{13}}{x_{A1,3}} = \frac{4}{7} = 0.5714$.

The normalisation results for all alternatives and criteria are presented in Table 4.

After the normalisation of the initial matrix, the next step of the MARCOS method involves the formation of the weighted normalised matrix. The weighted matrix is obtained by multiplying each element of the normalised matrix by the corresponding criterion weight w_j , where the weighting coefficients were previously determined using the FUCOM method ($w_1 = 0.236$, $w_2 = 0.063$, $w_3 = 0.132$, $w_4 = 0.184$, $w_5 = 0.154$, $w_6 = 0.119$, and $w_7 = 0.112$). In this way, the relative importance of each criterion is incorporated into the decision-making process, whereby criteria with higher weights have a greater influence on the final evaluation of alternatives. The results of calculating the weighted

normalised matrix are presented in Table 5. An example of calculating an element of the weighted normalised matrix is given below: $V_{11} = n_{11} \times w_1 = 0.3279 \times 0.236 = 0.0774$.

Table 3. Extended decision matrix

	C1	C2	C3	C4	C5	C6	C7
AAI	12.2	68,000	3	5	6	4	12
A1	9.5	12,800	4	5	6	7	60
A2	12.2	10,500	3	6	7	4	120
A3	4.0	68,000	7	7	7	6	12
A4	5.8	34,500	6	7	7	7	35
A5	6.4	24,800	7	7	7	7	90
AI	4.0	10,500	7	7	7	7	120
Optimisation objective	min	min	max	max	max	max	max

Note: A1 = direct road transport; A2 = rail transport with road distribution; A3 = direct air transport; A4 = combined road–air transport; A5 = integrated road–rail–air transport; AI = ideal alternative; AAI = anti-ideal alternative; C1 = delivery time; C2 = transport cost; C3 = transport network availability; C4 = reliability and resilience of the transport system; C5 = safety; C6 = flexibility; C7 = transport capacity. C1, C2, and C7 are quantitative criteria, while C3–C6 are qualitative criteria evaluated by experts using a seven-point Likert scale (1 = very low, 7 = very high). C1 is measured in hours (h), C2 in Bosnia and Herzegovina Convertible Marks (BAM), and C7 in tonnes (t).

Table 4. Normalised decision matrix

	C1	C2	C3	C4	C5	C6	C7
AAI	0.3279	0.1544	0.4286	0.7143	0.8571	0.5714	0.1000
A1	0.4211	0.8203	0.5714	0.7143	0.8571	1.0000	0.5000
A2	0.3279	1.0000	0.4286	0.8571	1.0000	0.5714	1.0000
A3	1.0000	0.1544	1.0000	1.0000	1.0000	0.8571	0.1000
A4	0.6897	0.3043	0.8571	1.0000	1.0000	1.0000	0.2917
A5	0.6250	0.4234	1.0000	1.0000	1.0000	1.0000	0.7500
AI	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Note: A1 = direct road transport; A2 = rail transport with road distribution; A3 = direct air transport; A4 = combined road–air transport; A5 = integrated road–rail–air transport; C1 = delivery time; C2 = transport cost; C3 = transport network availability; C4 = reliability and resilience of the transport system; C5 = safety; C6 = flexibility; C7 = transport capacity; AI = ideal alternative; AAI = anti-ideal alternative.

Table 5. Weighted normalised decision matrix

	C1	C2	C3	C4	C5	C6	C7
AAI	0.0774	0.0097	0.0566	0.1314	0.1320	0.0680	0.0112
A1	0.0994	0.0517	0.0754	0.1314	0.1320	0.1190	0.0560
A2	0.0774	0.0630	0.0566	0.1577	0.1540	0.0680	0.1120
A3	0.2360	0.0097	0.1320	0.1840	0.1540	0.1020	0.0112
A4	0.1628	0.0192	0.1131	0.1840	0.1540	0.1190	0.0327
A5	0.1475	0.0267	0.1320	0.1840	0.1540	0.1190	0.0840
AI	0.2360	0.0630	0.1320	0.1840	0.1540	0.1190	0.1120

Note: A1 = direct road transport; A2 = rail transport with road distribution; A3 = direct air transport; A4 = combined road–air transport; A5 = integrated road–rail–air transport; C1 = delivery time; C2 = transport cost; C3 = transport network availability; C4 = reliability and resilience of the transport system; C5 = safety; C6 = flexibility; C7 = transport capacity; AI = ideal alternative; AAI = anti-ideal alternative.

By applying Eq. (14), all values for the alternatives were summed as follows: $S_{AAI} = 0.0774 + 0.0097 + 0.0566 + 0.1314 + 0.1320 + 0.0680 + 0.0112 = 0.4863$. The remaining values were obtained in a similar manner. Subsequently, using Eq. (12), the utility degrees in relation to AAI were calculated as $K_1^- = \frac{S_1}{S_{AAI}} = \frac{0.6649}{0.4863} = 1.3673$. Using Eq. (13), the utility degree relative to AI was calculated as $K_1^+ = \frac{S_1}{S_{AI}} = \frac{0.6649}{1} = 0.6649$. The corresponding utility functions were then calculated using Eqs. (16) and (17): $f(K_1^-) = \frac{K_1^+}{K_1^+ + K_1^-} = \frac{0.6649}{0.6649 + 1.3673} = 0.3272$,

and $f(K_1^+) = \frac{K_1^-}{K_1^+ + K_1^-} = \frac{1.3673}{0.6649 + 1.3673} = 0.6728$. Finally, the utility function value for alternative A1 (direct road transport) was obtained using Eq. (15): $f(K_1) = 0.5736$. The remaining values were calculated in the same manner, and the final results are presented in Table 6.

Table 6. Results of MARCOS method

	S_i	K_i^-	K_i^+	$f(K_i^-)$	$f(K_i^+)$	$f(K_i)$	Rank
AAI	0.4863						
A1	0.6649	1.3673	0.6649	0.3272	0.6728	0.5736	5
A2	0.6887	1.4161	0.6887	0.3272	0.6728	0.5941	4
A3	0.8289	1.7045	0.8289	0.3272	0.6728	0.7151	2
A4	0.7847	1.6137	0.7847	0.3272	0.6728	0.6770	3
A5	0.8472	1.7421	0.8472	0.3272	0.6728	0.7309	1
AI	1.0000						

Note: MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; A1 = direct road transport; A2 = rail transport with road distribution; A3 = direct air transport; A4 = combined road–air transport; A5 = integrated road–rail–air transport; S_i = sum of the weighted normalised values for alternative A_i ; K_i^- = utility degree relative to the anti-ideal solution; K_i^+ = utility degree relative to the ideal solution; $f(K_i^-)$ = utility function relative to the anti-ideal solution; $f(K_i^+)$ = utility function relative to the ideal solution; $f(K_i)$ = final utility function of alternative A_i ; blank cells indicate non-applicable values.

The analysis of the results presented in Table 6 shows that alternative A5 (integrated road–rail–air transport) achieved the highest utility function value $f(K_5) = 0.7309$, resulting in its first-place ranking in the final assessment. The second-ranked alternative is A3 (direct air transport), with a utility function value of 0.7151, while the third position is occupied by A4 (combined road–air transport), with a value of 0.6770. These are followed by A2 (rail transport with road distribution), with a value of 0.5941, and A1 (direct road transport), with a value of 0.5736.

The obtained ranking indicates that A5 (integrated road–rail–air transport) represents the most favourable logistics solution for the distribution of humanitarian aid in the analysed scenario, taking into account the defined criteria and their weighting coefficients determined using FUCOM. At the same time, the differences in utility function values among the alternatives confirm that the application of the integrated FUCOM–MARCOS model enables clear differentiation of the quality of the considered transport solutions and provides a reliable basis for selecting the optimal alternative in MCDM conditions.

6 Sensitivity Analysis

To assess the stability of the proposed FUCOM–MARCOS model, a sensitivity analysis was carried out to examine the effect of changes in criterion weights on the final ranking of the considered alternatives. The weight of each criterion was gradually reduced to 95%, 85%, 75%, 65%, 55%, 45%, 35%, 25%, 15%, and 5% of its initial value, while the initial weights of the other criteria were retained. In each scenario, the resulting criterion weights were subsequently renormalised so that their sum was equal to 1. In total, 70 scenarios (S1–S70) were developed, and the MARCOS method was reapplied for each scenario to determine the resulting ranking of the alternatives. Figure 1 provides a comparative overview of the changes in the rankings of all alternatives across the analysed scenarios.

The sensitivity analysis results indicate that the model remained highly stable across most of the analysed scenarios. Alternative A5 (integrated road–rail–air transport), which was ranked as the best solution in the baseline model, remained in first place in nearly all scenarios, with only a few cases where it was overtaken by A3 (direct air transport). Alternative A3 (direct air transport) also generally retained second place, although it occasionally moved to the first position. This suggests that A3 (direct air transport) and A5 (integrated road–rail–air transport) are the most competitive alternatives and have relatively similar overall performance.

Alternative A4 (combined road–air transport) showed the greatest stability among the medium-ranked alternatives, remaining in third place in most of the analysed scenarios, with only occasional changes in rank. Alternative A2 (rail transport with road distribution) was also relatively stable, generally occupying fourth place, whereas A1 (direct road transport) remained in fifth place in most scenarios. In a small number of scenarios, A1 (direct road transport) and A2 (rail transport with road distribution) exchanged their relative positions, indicating that their overall utility values are relatively close. Consequently, relatively small changes in the weights of individual criteria may alter the relative ranking of these two alternatives, although their overall positions in the lower part of the ranking remain largely stable.

Overall, the sensitivity analysis confirms that changes in the criterion weights have little effect on the final ranking of the alternatives. The results therefore demonstrate the robustness and stability of the FUCOM–MARCOS model and support the reliability of the obtained results in selecting the most suitable transport alternative for delivering humanitarian aid in natural disaster conditions.

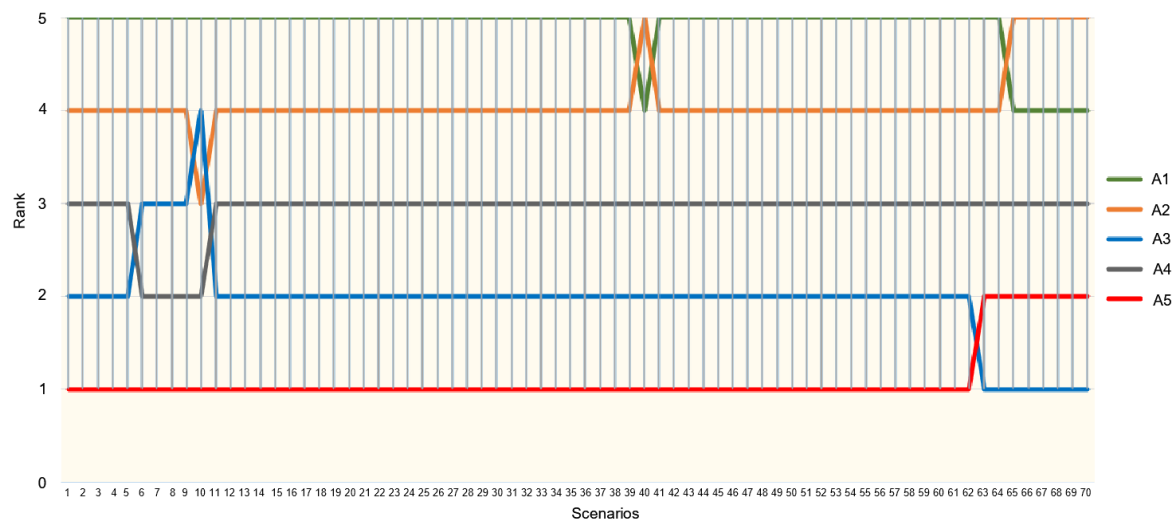


Figure 1. MARCOS-based ranking stability of multimodal transport alternatives across 70 criterion-weight sensitivity scenarios

Note: MARCOS = Measurement of Alternatives and Ranking according to Compromise Solution; A1 = direct road transport; A2 = rail transport with road distribution; A3 = direct air transport; A4 = combined road–air transport; A5 = integrated road–rail–air transport.

7 Conclusion

Selecting a transport solution for crisis and humanitarian operations requires decision-makers to balance multiple operational, economic, safety, and resilience-related criteria. The problem becomes more difficult when natural disasters damage transport infrastructure, reduce network capacity, or make established routes unavailable. This study addressed this problem by applying an integrated FUCOM–MARCOS model to identify the most suitable multimodal transport arrangement for humanitarian aid delivery in Bosnia and Herzegovina.

FUCOM was used to determine the relative weights of seven criteria describing the operational, economic, and organisational characteristics of the transport alternatives. Delivery time (C1) received the highest weight of 0.236, followed by transport system reliability and resilience (C4) at 0.184 and safety (C5) at 0.154, whereas transport cost (C2) received the lowest weight of 0.063. This weighting structure indicates that response speed, continuity of transport operations, and delivery safety take precedence over cost considerations in the examined humanitarian setting.

MARCOS was then used to evaluate five alternatives for transporting 60 tonnes of humanitarian aid from the Armed Forces logistics base at Rajlovac Barracks to the Donja Jablanica area following the floods and landslides of October 2024. The A5 (integrated road–rail–air transport) achieved the highest utility value (0.7309), followed closely by A3 (direct air transport), with a utility value of 0.7151. The result suggests that combining transport modes is the most suitable arrangement for the conditions represented in the case study. Its first-place ranking reflects its overall performance across delivery time, network availability, reliability, safety, flexibility, capacity, and cost rather than superiority against every individual criterion.

The sensitivity analysis examined 70 scenarios in which the criterion weights were varied. The leading alternative retained first place in nearly all scenarios, although direct air transport ranked first in a small number of cases. The remaining alternatives also showed limited changes in position. These results indicate that the baseline ranking was generally stable, while the small difference between A5 (integrated road–rail–air transport) and A3 (direct air transport) shows that the preferred solution may change when decision priorities shift. Sensitivity analysis therefore provides important information for emergency planners rather than merely serving as a statistical confirmation of the initial ranking.

The study contributes to research on humanitarian logistics and disaster-resilient transport planning by framing multimodal transport selection as a decision problem under infrastructure disruption. Methodologically, it connects consistent expert-based criterion weighting with compromise-based alternative ranking and tests the stability of the resulting order. From an urban and regional management perspective, the framework shows how the availability, reliability, and redundancy of transport infrastructure can be incorporated into emergency logistics decisions. It provides a transparent decision-support basis for the Ministry of Defence of Bosnia and Herzegovina, the Armed Forces of Bosnia and Herzegovina, civil protection and disaster management institutions, humanitarian organisations, and other bodies responsible for maintaining access to disaster-affected communities.

The findings should nevertheless be interpreted within the boundaries of the research design. The model was

applied to one representative logistics scenario, five predefined transport alternatives, and a specific origin-destination pair in Bosnia and Herzegovina. Some input values were derived from the assessments of five experts and may therefore reflect the experience and priorities of this group. Moreover, the model produced a comparative ranking for the examined scenario; it did not simulate dynamic network conditions, route-level disruptions, transfer delays, or changes in aid demand during the operation. The results should consequently not be treated as a universally optimal transport plan for all disasters or geographical settings.

Future research may examine additional disaster types, origin–destination configurations, transport alternatives, and evaluation criteria. Fuzzy or probabilistic approaches could be used to represent uncertainty in expert assessments and infrastructure conditions, while comparisons with other MCDM methods could test the sensitivity of the ranking to the selected analytical procedure. Integrating the framework with geographic information systems, network simulation, and real-time information on route availability, weather, transport capacity, and aid demand could support dynamic decision-making during emergency operations. Such extensions would move the model from scenario-based alternative selection towards an operational decision-support system for disaster-resilient humanitarian transport planning.

Author Contributions

Conceptualization, D.B. and M.B.; methodology, D.B. and Ž.S.; validation, Ž.S.; formal analysis, D.B.; data curation, D.B.; writing—original draft preparation, D.B. and M.B.; writing—review and editing, Ž.S.; supervision, Ž.S. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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